

Lab 1. EE – Multi-armed bandit

Multi-armed bandit (MAB) framework has attracted a lot of attention in various applications, from recommender systems and information retrieval to healthcare and finance, due to its stellar performance combined with certain attractive properties, such as learning from less feedback. The multi-armed bandit field is currently flourishing, as novel problem settings and algorithms motivated by various practical applications are being introduced.

In this lab you will be asked to experiment with MAB in stationary and non-stationary settings.

If you are not familiar with Google Colab, follow first [this tutorial](#).

1. Copy the Python source code provided in Atenea into a Google Drive folder and open it in Google Colab. Read the code in the notebook, run it, and analyze the results (average reward obtained in convergence, evolution of the percentage of correct decisions and the estimated value of $Q(a)$ for every action on a single run).
2. Use the Python code provided as a basis to include your own code and solve the next questions:
 - a) For ε -greedy algorithm, check the reward on a single run for smaller values of the variance of the Gaussian $f(r|a)$ in example 2.1 in slides. Modify the code to compute the regret over time. Derive conclusions.
 - b) Program the UCB technique. Check the average reward obtained in convergence for several values of the parameter and the evolution of the percentage of correct decisions. Check also the estimated value of $Q(a)$ for every action on a single run and comment.
 - c) Plot the evolution of regret for ε -greedy and UCB algorithms, compare them and comment.
 - d) Think of a practical application that can be modeled with an m-armed bandit. Use a sensible $f(r|a) \forall a$ (Gaussian, binary Bernoulli, exponential, etc.) for that application. You may get inspiration from [this paper](#). Assume **stationarity** over time. Program it using the base code provided. Note that if Gaussian rewards cannot be assumed in your application, you'll have to do the proper modifications.
 - e) Program another MAB procedure among those reviewed in the theory lectures (gradient bandit, softmax or probability matching) and compare performance to the provided by ε -greedy and UCB in the practical application given in d.
 - f) Make the environment in d) **non-stationary**, and check the performance of the procedures ε -greedy, UCB and the provided in e) in this case.
 - g) Choose one of the algorithms in section e) and include the appropriate changes in the estimation of $Q(a)$ to allow the algorithm track the best decisions in the non-stationary environment.
3. Prepare a report in Colab containing a description and interpretation of results. Remember to give the appropriate access permit to the instructors.
4. Hand in a link to your report in Atenea.